

Quiz on model graphs

* Required

* This form will record your name, please fill your name.

Instructions

Instructions:

1. Each MCQ carries marks as given across the question.
2. Each MCQ may have more than one or no options.
- 3.

$$\text{sgn}(q) = \begin{cases} +1 & q > 0 \\ -1 & q < 0 \end{cases}$$

4. Every registration number is mapped to a roll number is R as given in attendance sheet.
5. To decide which option to choose at layer n of the question, first write the roll number in the binary representation. For example, 27 in base 10 representation is equivalent to 011011 in base 2 representation. 7 is the least significant digit in the base 10 representation, whereas 1 is the least significant bit in the base 2 representation. Similarly 2 is the most significant digit in the base 10 representation, whereas 0 is the most significant digit in the base 2 representation.

1

Choose place 0 bit *

Layer 0 – Thermistor

Problem. To measure the temperature coefficient α in a material, we measure the resistance R vs temperature T . The guiding equation is

$$\alpha = \frac{R(T_2) - R(T_1)}{R(T_1)} \frac{1}{T_2 - T_1}$$

☐ 0

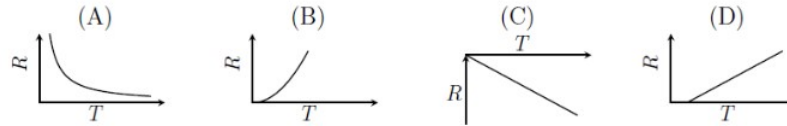
☐ 1

Layer 0 - bit 0

2

* (1 Point)

L0i Which of the following model graphs best represents the dependence of resistance R on the temperature T for an extrinsic semiconductor?



☐ A

☐ B

☐ C

☐ D

3

* (1 Point)

L0ii For negative temperature coefficient (NTC) type thermistor, how is the current flowing through it is related to temperature?

(A) directly (B) indirectly (C) independent (D) none of the above

☐ A

☐ B

☐ C

☐ D

4

* (1 Point)

L0iii Resistance of thermistor is $[(R\%2) + 1] \text{ k}\Omega$ @ 35°C and $[(R\%2) + 0.5] \text{ k}\Omega$ @ 45°C . The temperature coefficient @ 35°C is given by

$$\begin{array}{cccc} \text{(A)} & \text{(B)} & \text{(C)} & \text{(D)} \\ -1 \times 10^{-1} ^\circ\text{C}^{-1} & 1 \times 10^{-1} ^\circ\text{C}^{-1} & -5 \times 10^{-2} ^\circ\text{C}^{-1} & 5 \times 10^{-2} ^\circ\text{C}^{-1} \end{array}$$

☐ A

☐ B

☐ C

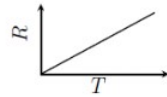
☐ D

Layer 0 - bit 1

5

* (1 Point)

L0i Resistance vs temperature of a material was found to be



The material is likely to be

- (A) metal (B) intrinsic semiconductor (C) extrinsic semiconductor (D) none of the above

☐ A

☐ B

☐ C

☐ D

6

* (1 Point)

L0ii For positive temperature coefficient (PTC) type thermistor, how is the current flowing through it is related to temperature?

(A) directly (B) indirectly (C) independent (D) none of the above

☐ A

☐ B

☐ C

☐ D

7

* (1 Point)

L0iii Resistance of thermistor is $[(R\%2) + 0.5] \text{ k}\Omega$ @ 35°C and $[(R\%2) + 1] \text{ k}\Omega$ @ 45°C . The temperature coefficient @ 35°C is given by

$$\begin{array}{cccc} \text{(A)} & \text{(B)} & \text{(C)} & \text{(D)} \\ -1 \times 10^{-1} ^\circ\text{C}^{-1} & 1 \times 10^{-1} ^\circ\text{C}^{-1} & -5 \times 10^{-2} ^\circ\text{C}^{-1} & 5 \times 10^{-2} ^\circ\text{C}^{-1} \end{array}$$

☐ A

☐ B

☐ C

☐ D

Layer 1 - Photo electric effect

Problem. To measure the photoelectric effect in Au, we measure the stopping potential V_{sp} vs photon frequency ν to estimate the work function Φ and threshold frequency ν_0 . The separation of light source and vacuum phototube determines the intensity of light I . The guiding equation is

$$eV_{sp} = h\nu - \Phi$$

8

Choose place 1 bit *

☐ 0

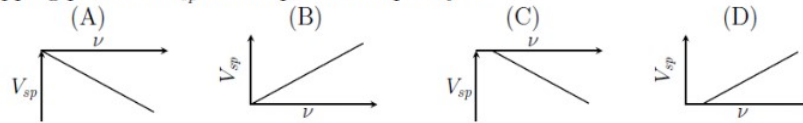
☐ 1

Layer 1 - bit 0

9

* (1 Point)

Li stopping potential V_{sp} on the photon frequency ν ?



☐ A

☐ B

☐ C

☐ D

10

* (1 Point)

L1ii For an electromagnetic wave, how is its frequency related to its wavelength?

(A)	(B)	(C)	(D)
directly	indirectly	independent	none of the above

☐ A

☐ B

☐ C

☐ D

11

* (1 Point)

L1iii Work function of a photosensitive material is $[(R\%2) + 1]$ eV. The energy of photon incident is 3 eV. The kinetic energy of photo-electron is

(A)	(B)	(C)	(D)
0 eV	1 eV	2 eV	3 eV

☐ A

☐ B

☐ C

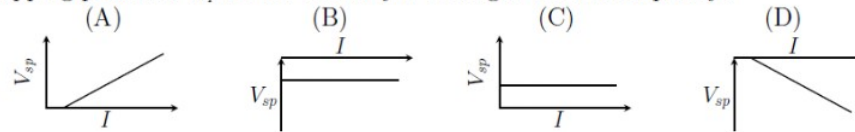
☐ D

Layer 1 bit 1

12

* (1 Point)

L1i stopping potential V_{sp} on the intensity I of a light of fixed frequency?



☐ A

☐ B

☐ C

☐ D

13

* (1 Point)

L1ii For an electromagnetic wave, how is its amplitude related to its intensity?

(A) directly (B) indirectly (C) independent (D) none of the above

☐ A

☐ B

☐ C

☐ D

14

* (1 Point)

L1iii Work function of a photosensitive material is 3 eV. . The energy of photon incident is $[(R\%2) + 3]$ eV. The kinetic energy of photo-electron is

(A)	(B)	(C)	(D)
0 eV	1 eV	2 eV	3 eV

☐ A

☐ B

☐ C

☐ D

Layer 2 - Energy band gap of semiconductor

Problem. To measure the energy band gap of Germanium intrinsic semiconductor, we measure the current with varying temperature. The guiding equation is

$$\sigma = \sigma_0 \exp \left(- \frac{E_g}{2k_B T} \right)$$

[Hint For the following analysis, take the logarithm of above equation.]

15

Choose place 2 bit *

☐ 0

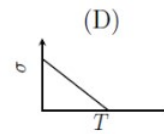
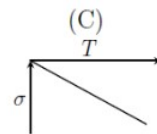
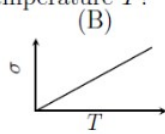
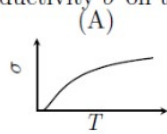
☐ 1

Layer 2 - bit 0

16

* (1 Point)

L2i conductivity σ on the temperature T ?



☐ A

☐ B

☐ C

☐ D

17

* (1 Point)

L2ii For an extrinsic semiconductor, how is doping concentration related to its energy band gap?

(A) directly (B) indirectly (C) independent (D) none of the above

☐ A

☐ B

☐ C

☐ D

18

* (1 Point)

L2iii The slope of $\frac{1000}{T}$ vs $\log_{10} I$ plot is $[(R\%2)+2]$ K. The energy band gap of semiconductor is

(A) 0.8 eV (B) 1.2 eV (C) 0.4 eV (D) 1.6 eV

☐ A

☐ B

☐ C

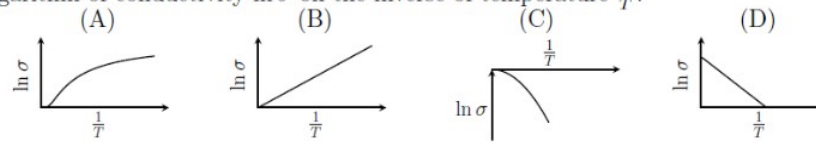
☐ D

Layer 2 - bit 1

19

* (1 Point)

L2i logarithm of conductivity $\ln \sigma$ on the inverse of temperature $\frac{1}{T}$?



☐ A

☐ B

☐ C

☐ D

20

* (2 Points)

L2ii For a semiconductor, how is its resistance related to its temperature?

(A) directly (B) indirectly (C) independent (D) none of the above

☐ A

☐ B

☐ C

☐ D

21

* (1 Point)

L2iii The energy band gap of semiconductor is $0.4 \cdot [(R\%2)+2]$ eV. The slope of $\frac{1000}{T}$ vs $\log_{10} I$ plot is .

(A)	(B)	(C)	(D)
1 K	2 K	3 K	4 K

☐ A

☐ B

☐ C

☐ D

Layer 3 - Hall effect

Problem. A thin conducting foil of fixed width and length carries a steady current I and is placed in a uniform magnetic field $\vec{B}$ perpendicular to the plane of the foil. The transverse (Hall) voltage is measured between points on opposite faces across the thickness of the foil while the current and magnetic field are held constant.

22

Choose place 3 bit *

☐ 0

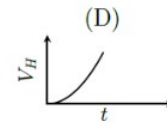
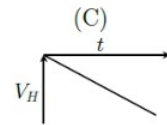
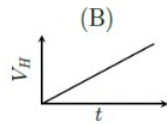
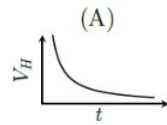
☐ 1

Layer 3 - bit 0

23

* (1 Point)

L3i Hall voltage V_H on the foil thickness t ?



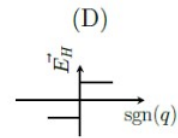
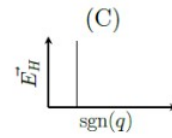
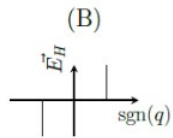
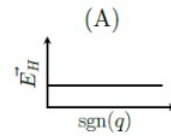
☐ A

☐ B

☐ C

☐ D

* (1 Point)

L3ii Hall field $\vec{E}_H$ on the sign of carriers $\text{sgn}(q)$?
☐ A

☐ B

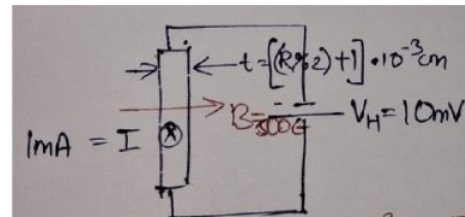
☐ C

☐ D

25

* (1 Point)

L3iii Current passing through the semiconductor crystal is 1 mA. The thickness of the crystal is $[(R/2) + 1] \cdot 10^{-3}$ cm. In the presence of magnetic field of 500 G, the Hall voltage generated is 10 mV. R_H of semiconductor is



- (A) $1 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (B) $2 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (C) $3 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (D) $4 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$

☐ A

☐ B

☐ C

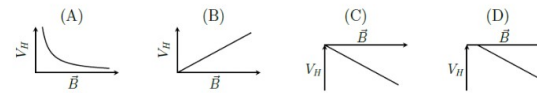
☐ D

Layer 3 - bit 1

26

* (2 Points)

L3i Hall voltage V_H on the magnetic field $\vec{B}$?



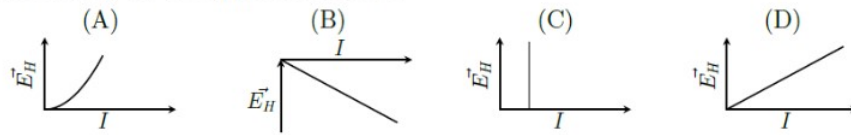
☐ A

☐ B

☐ C

☐ D

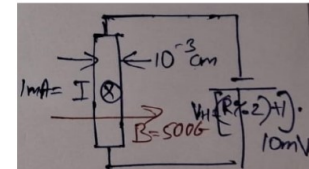
* (2 Points)

L3ii Hall field E_H on the applied current I ?☐ A☐ B☐ C☐ D

28

* (1 Point)

L3iii Current passing through the semiconductor crystal is 1 mA. The thickness of the crystal is 10^{-3} cm. In the presence of magnetic field of 500 G, the Hall voltage generated is $[(R_H/2) + 1] \cdot 10$ mV. R_H of semiconductor is



- (A) $1 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (B) $2 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (C) $3 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$ (D) $4 \times 10^{-3} \text{ m}^3 \text{ C}^{-1}$

☐ A

☐ B

☐ C

☐ D

Layer 4 - Composition of EBG and Hall

29

Choose place 4 bit *

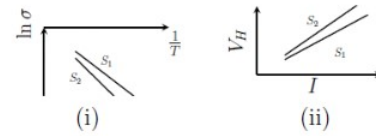
☐ 0

☐ 1

Layer 4 - bit 0

* (2 Points)

L4i The conductivity vs temperature and Hall voltage vs current profiles for two semiconductors S_1, S_2 is given below. The energy band gaps are given respectively by E_1, E_2 .



If the truth value of the statement $E_1 > E_2$ is T_1 , and truth value of graph (ii) is T_2 , then the truth value pair (T_1, T_2) is given by

(A) (T, T) (B) (T, F) (C) (F, T) (D) (F, T)

☐ A

☐ B

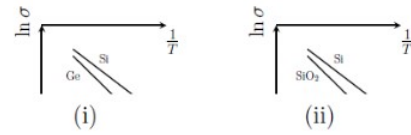
☐ C

☐ D

Layer 4 - bit 1

* (2 Points)

L4i The conductivity vs temperature profiles for three materials Si, Ge, SiO_2 is given below.
If the truth value of graph (i) is T_1 and the truth value of graph (ii) is T_2 , then the



truth value pair (T_1, T_2) is given by

(A) (T, T) (B) (T, F) (C) (F, T) (D) (F, T)

☐ A

☐ B

☐ C

☐ D

Layer 5 - Composition of Metals and semiconductors

32

Choose place 5 bit *

☐ 0

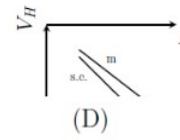
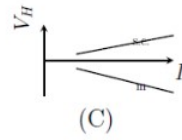
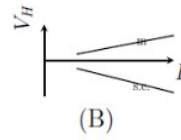
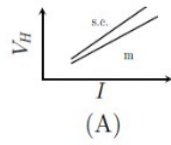
☐ 1

Layer 5 - bit 0

33

* (2 Points)

L5i Choose the correct model graph(s).



☐ A

☐ B

☐ C

☐ D

Layer 5 - bit 1

34

* (2 Points)

L5i Choose the correct model graph(s).

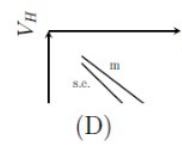
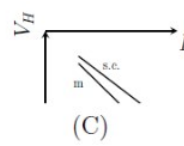
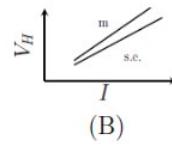
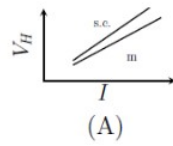


Table 1: m is for metal; s.c. is for semiconductor.

☐ A

☐ B

☐ C

☐ D

